

Vine Copula VAR

From Recursive Margins to Joint Forecast Inference

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Time-series forecasts and joint events

Motivation



- Time-series forecasts use past observations to predict what comes next.
- Several series can move together. Separate forecasts do not tell us the chance of a joint event ([Patton, 2012](#)).
- Because the model is fitted to limited past data, its reported joint probability is only an estimate. We need to measure how precise that estimate is.

Model



Findings



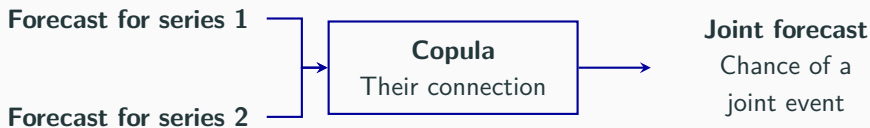
Our aim: a joint-event forecast with a confidence interval for its estimated probability.

Conclusion



What a copula adds

- Each individual forecast describes what could happen to one series.
- A copula describes how their surprises move together, including whether weak surprises tend to arrive together ([Patton, 2012](#)).
- Combining the individual forecasts with that connection gives a joint forecast.



Motivation



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Current literature: what is still unknown

- **What they do.** Nagler et al. (2022) model multivariate time-series dependence with vines and quantify uncertainty in predictions. They use one fitted distribution per series for the full history. Newer work develops richer or larger models (Chen et al., 2025; Gauss and Nagler, 2026).
- **What is not known.** How precisely can we estimate today's joint-event probability when we learn from past forecasts made with different estimates at different dates?

Motivation



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Proposed model: VAR forecasts with vine dependence

Time dynamics and individual forecast surprises

$$\mu_t = a + \sum_{\ell=1}^p A_{\ell} y_{t-\ell}, \quad y_t = \mu_t + D_{\sigma} Z_t, \quad Z_t \text{ iid}, \quad Z_{it} \sim N(0, 1).$$

Their connection: a three-variable vine (Aas et al., 2009)

$$U_{it} = \Phi(Z_{it}), \quad c_{\theta}(u) = c_{12}(u_1, u_2) c_{23}(u_2, u_3) c_{13|2}(u_{1|2}, u_{3|2}).$$

$u_{1|2}$ and $u_{3|2}$ are conditional percentiles given series 2; c is a density.

Next joint-event probability

$$\hat{p}_{T+1}(q) = C_{\hat{\theta}} \left(\left[\Phi \left(\frac{q_i - \hat{\mu}_{i,T+1}}{\hat{\sigma}_{i,T}} \right) \right]_{i=1}^3 \right).$$

q_i : event threshold; $\hat{\mu}_{i,T+1}$: current VAR mean forecast; C : copula CDF. $D_{\sigma} = \text{diag}(\sigma_1, \sigma_2, \sigma_3)$. Stable VAR, constant coefficients and scales, fixed correctly specified simplified vine.

Fit the vine to dated surprises and forecast with today's margins. We now explain the steps.

Motivation



Model



Findings

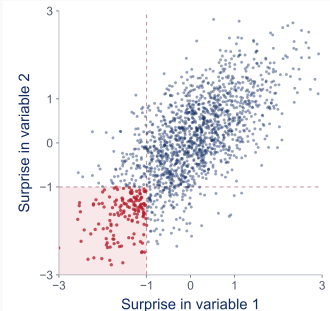


Conclusion



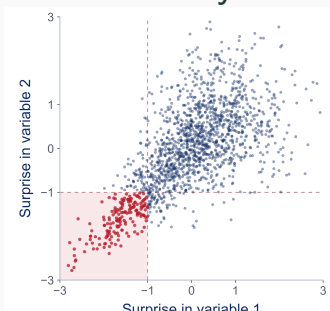
Gaussian and Clayton pairs inside a vine

Gaussian



Symmetric dependence.

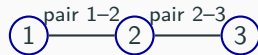
Positive Clayton



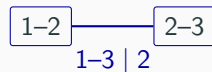
Extreme lows cluster.

Illustrative draws. Same normal margins and rank dependence; red marks two low surprises.

Three-variable vine



Then connect 1 and 3
after accounting for 2



Pairs and conditional pairs.

Motivation



Model



Findings



Conclusion



A vine combines these pairs (Aas et al., 2009; Czado and Nagler, 2022). We fit it to dated forecast surprises.

From issued forecasts to the next joint probability

$$\begin{aligned}\hat{U}_{it} &= \hat{F}_{i,t|t-1}(y_{it}), & \hat{u}_{i,T+1} &= \hat{F}_{i,T+1|T}(q_i). \\ \hat{p}_{T+1} &= C_{\hat{\theta}}(\hat{u}_{1,T+1}, \dots, \hat{u}_{n,T+1}).\end{aligned}$$

- Before each outcome arrives, fit the VAR using earlier observations.
- Record its issued forecast percentile, or PIT, then update the VAR (Diebold et al., 1998; Dawid, 1984).
- Fit the vine to these saved PITs. Combine it with today's threshold probabilities to forecast the next joint event.

We now have a probability forecast. Its estimated inputs also create uncertainty about that probability.

Motivation



Model



Findings



Conclusion



Where the probability forecast gets its uncertainty

$$\hat{p}_{T+1} = p(\underbrace{\hat{\beta}_T, \hat{\sigma}_T}_{\text{current VAR}}, \underbrace{\hat{\theta}_T}_{\text{vine from dated PITs}}; x_{T+1}, q).$$

$x_{T+1} = (1, y'_T, \dots, y'_{T-p+1})'$ is today's observed lag state; q fixes the event thresholds.

- Errors in today's VAR estimates affect the individual threshold probabilities.
- The vine estimate inherits uncertainty from past marginal fits and its own pair fits (Hobæk Haff, 2013).
- The estimates share data. We combine their errors and cross covariances in one standard error (Murphy and Topel, 1985).

The findings first turn this standard error into an interval, then check its size and coverage.

Motivation



Model



Findings



Conclusion



Theory: an interval for the joint probability

$$\hat{p}_{T+1} \pm 1.96 \widehat{SE}(\hat{p}_{T+1})$$

- The interval measures uncertainty about today's **conditional joint-event probability**. Random-state inference requires care ([Beutner et al., 2021](#)).
- Across repeated samples, coverage approaches 95% under our model and regularity assumptions.
- A logit interval works on log odds, then returns to probabilities with endpoints between zero and one.

This interval combines the estimation errors. We then ask what keeping dated forecasts adds to its variance.

Motivation



Model



Findings



Conclusion



The variance cost of preserving issued margins

- A common marginal fit for historical transforms is a natural benchmark (Nagler et al., 2022).
- Our terminal refit recomputes old transforms using current estimates, holding the observations fixed.
- In the **Gaussian submodel**, the archive adds nonnegative asymptotic event variance.

$$\text{archive variance} = \text{same-history refit variance} + \text{archive cost}$$

This same-history result is asymptotic; finite-sample SEs need not be ordered. We check interval coverage next in simulations.

Motivation



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Conclusion



Simulation check: lower-tail interval coverage

Three simulated time series, persistent VAR(1)

Each training date supplies all three values; T counts these dates.

Nominal 95% intervals for all three outcomes ≤ -1 ; 1,000 datasets per design.

Training dates T	Gaussian vine		Clayton vine	
	Wald	Logit	Wald	Logit
200	85.8%	95.0%	86.8%	94.5%
500	87.5%	93.8%	91.9%	94.8%
800	90.9%	94.4%	92.7%	94.7%

- Coverage is the share of 1,000 intervals containing the true joint-event probability.
- With the same forecast and SE, logit is closer to 95% than Wald in each row.

These simulations check inference. We next benchmark forecasts on observed releases.

Motivation



Model



Findings



Conclusion



Benchmarking: a three-variable VAR(1)

$$y_m = (10E_m, I_m, -10U_m)', \quad x_{m+1} = (1, y_{1m}, y_{2m}, y_{3m})'.$$

E, I : employment and production growth; U : unemployment-rate change.

- **3 forecast variables** from [ALFRED](#) vintages: PAYEMS, INDPRO and UNRATE.
- **4 entries in x per equation**: the three lagged variables plus an intercept. There are no extra external predictors.
- **390 forecasts (July 1987–December 2019)**, plus 60 stress forecasts (2020–2024).

Labor contraction	Employment falls and unemployment rises
Broad contraction	Employment and production fall, unemployment rises
Employment–production	Employment and production both fall

All rules predict the same locked releases (Clements and Galvão, 2023). We compare their Brier losses next.

Motivation



Model



Findings



Conclusion



Benchmarking: prediction by Brier loss

Labor-contraction forecasts. Smaller Brier loss is better.

Model	1987–2019	2020–2024
Archived Gaussian	0.0882	0.1448
Locked-history Gaussian refit	0.0872	0.1783
Archived Clayton	0.0871	0.1011
Independence benchmark	0.0834	0.0996

- Brier loss averages squared gaps between probabilities and zero-or-one outcomes (Gneiting and Raftery, 2007).
- Independence has lower loss than either archive in both periods.

This descriptive comparison has no significance ranking. Prediction and estimation uncertainty differ; we examine the standard error next.

Motivation



Model



Findings



Conclusion



Benchmarking: what drives the standard error

Primary-period archived Gaussian forecasts

Median diagnostic	Labor	Employment–production
Complete standard error	0.0178	0.0177
Extra archive term / total variance	1.86%	2.89%
SE omitting current margins / full SE	37.7%	32.1%

- Omitting current-margin uncertainty makes the reported SE much smaller.
- Serial dependence and changing error scales question our constant-parameter margins. Richer copula VARs allow parameter changes ([Tsionas et al., 2022](#)).

Current margins dominate the uncertainty adjustment here. The intervals remain model based; coverage for the release process is unestablished.

Motivation



Model



Findings



Conclusion



- **Gap.** Past forecasts and today's forecast use different estimates based on shared data. Existing common-fit uncertainty formulas do not handle this.
- **Proposal.** Use a Vine Copula VAR to combine individual forecasts through a vine copula. Our confidence intervals include linked errors from dated marginal fits, current VAR estimates, and vine estimation.
- **Simulation result.** Logit beats Wald on closeness to 95% lower-tail coverage in the rows shown: **95.0% versus 85.8%** at 200 training dates for Gaussian.
- **Benchmarking result.** Current VAR uncertainty matters most. Archived forecasts have no consistent Brier-loss advantage over refitting.

Motivation



Model



Findings



Conclusion



B1. Maintained model and scope

$$y_t = a + \sum_{j=1}^p A_j y_{t-j} + D_\sigma Z_t, \quad \rho(\mathcal{A}) < 1, \quad Z_{it} \sim N(0, 1).$$

- Z_t is iid across dates. Dependence within a date follows a fixed, correctly specified simplified vine (Czado and Nagler, 2022).
- Dimension, lag order, coefficients and scales remain fixed.
- Gaussian and positive Clayton pairs satisfy the paper's regularity conditions. Free parameters are interior.
- Selection, boundary estimates and estimated thresholds need further analysis.

A dated record preserves timing. These assumptions still need checking.

B2. A three-variable vine and sequential fitting

$$c(u_1, u_2, u_3) = c_{12}(u_1, u_2)c_{23}(u_2, u_3)c_{13|2}(u_{1|2}, u_{3|2}).$$

$$u_{1|2} = \partial_{u_2} C_{12}(u_1, u_2), \quad u_{3|2} = \partial_{u_2} C_{23}(u_2, u_3).$$

- Fit pairs (1, 2) and (2, 3) first, then the conditional pair (1, 3 | 2) ([Aas et al., 2009](#)).
- The conditional inputs depend on earlier estimates, so their errors enter the later score ([Hobæk Haff, 2013](#)).
- The simplifying assumption holds the conditional pair parameter constant across conditioning values.

B3. Historical overlap creates harmonic weights

$$h_{T,s} = \sum_{m=\max(b_T,s)}^{T-1} \frac{1}{m}, \quad b_T = \lceil T^\eta \rceil, \quad 0 < \eta < 1.$$

- Innovation s affects many later marginal fits. Its weight in a fit of size m is proportional to $1/m$.
- Earlier innovations enter more fits and receive larger cumulative weights, as in recursive forecast inference ([West, 1996](#)).
- The intercept identity $\mathbb{E}(x_t)'Q^{-1}x_s = 1$, with $Q = \mathbb{E}(x_t x_t')$, and stability simplify the historical correction.

Historical location and scale moments remain. Every current slope still enters the forecast at its single realized lag state.

B4. The joint influence and covariance

$$\sqrt{T}(\hat{\theta} - \theta_0) = -\frac{J^{-1}}{\sqrt{T}} \sum_{s=1}^T \{\mathbf{1}_{\{s > b_T\}} g_s + h_{T,s} B \varphi_s\} + o_p(1).$$

$$\xi_t = \begin{pmatrix} -J^{-1}(g_t + B\varphi_t) \\ e_t \end{pmatrix}, \quad \mathcal{W} = \mathbb{E}(\xi_t \xi_t') + \begin{pmatrix} J^{-1} B \Sigma_{\varphi} B' J^{-1} & 0 \\ 0 & 0 \end{pmatrix}.$$

- g_t : stacked vine scores. J : their parameter Jacobian. B : sensitivity to marginal location and variance errors.
- $\varphi_t = (Z_t', Z_{1t}^2 - 1, \dots, Z_{nt}^2 - 1)'$ and $\Sigma_{\varphi} = \mathbb{E}(\varphi_t \varphi_t')$.
- e_t : terminal coefficient and relative-variance influences. The common outer product retains cross covariances. This shared-error issue also underlies two-stage inference (Murphy and Topel, 1985).

B5. Event gradients and the random forecast state

$$z_i(r) = \frac{c_i - r' \beta_i}{\sigma_i}, \quad u_i(r) = \Phi\{z_i(r)\}, \quad \tau^2(r) = d(r)' \mathcal{W} d(r).$$

$$d_\theta(r) = \partial_\theta C_{\theta_0}\{u(r)\},$$

$$d_{\beta_i}(r) = -C_i \phi\{z_i(r)\} r, \quad d_{v_i}(r) = -\frac{1}{2} C_i z_i(r) \phi\{z_i(r)\}.$$

Gradients use normalized coefficient and relative-variance coordinates. $C_i = \partial C / \partial u_i$.

- The coefficient gradient uses every component of the realized state r .
- Under stability, full-history estimation error separates asymptotically from the recent innovations that determine the state ([Beutner et al., 2021](#), related sample-separation principle:).
- The state-dependent standard error $\hat{\tau}_T / \sqrt{T}$ yields a normal pivot and repeated-sample coverage.

B6. Confidence intervals on log odds

$$\ell(p) = \log \frac{p}{1-p}, \quad \Lambda(z) = \frac{1}{1 + e^{-z}}.$$

$$I_\ell = \left[\Lambda \left\{ \ell(\hat{p}) - \frac{1.96\hat{s}}{\hat{p}(1-\hat{p})} \right\}, \quad \Lambda \left\{ \ell(\hat{p}) + \frac{1.96\hat{s}}{\hat{p}(1-\hat{p})} \right\} \right], \quad \hat{s} = \frac{\hat{\tau}_T}{\sqrt{T}}.$$

- The delta method gives $\text{SE}\{\ell(\hat{p})\} \approx \hat{s}/[\hat{p}(1-\hat{p})]$.
- Returning to probability gives asymmetric endpoints inside $(0, 1)$.
- The first-order coverage result is preserved for interior probabilities and positive event variance.

B7. Gaussian archive cost and event sensitivity

$$\mathcal{W}_{\text{ar}} - \mathcal{W}_{\text{refit}} = \text{diag}\{J^{-1}B\Sigma_{\varphi}B'J^{-'}, 0\} \succeq 0.$$

$$\tau_{\text{ar},A}^2(r) = \tau_{\text{refit},A}^2(r) + q_A(r), \quad q_A(r) \geq 0.$$

Bivariate lower-orthant case: $q_A(r) = \phi_2\{z_1(r), z_2(r); \rho\}^2 \frac{\rho^2(1 - \rho^2)^2}{1 + \rho^2}.$

- For the same observations and Gaussian limiting covariances, the extra cost vanishes at independence and depends on the event's sensitivity to correlation.
- Its share can be small because current marginal uncertainty is large.
- lid zero-threshold example: at $\rho = 0.5$, leverage rising from 1 to 3 reduces the share from 2.67% to 1.00%.

B8. Exact weights for the leading finite-sample array

$$b = b_T, \quad N = T - b, \quad H_T = \sum_{m=b}^{T-1} \frac{1}{m}.$$

$$k_{11} = T/N, \quad k_{12} = \frac{T(N - bH_T)}{N^2}, \quad k_{22} = \frac{T\{2N - (2b - 1)H_T\}}{N^2}.$$

- These replace limiting coefficients 1, 1, 2 on score, cross and historical covariance terms.
- At $T = 500$, $b = \lceil T^{0.7} \rceil = 78$, they are 1.1848, 0.7768 and 1.5588.
- Terminal blocks are unchanged. The estimator remains positive semidefinite and consistent.

These exact leading-array weights do not guarantee finite-sample coverage.

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